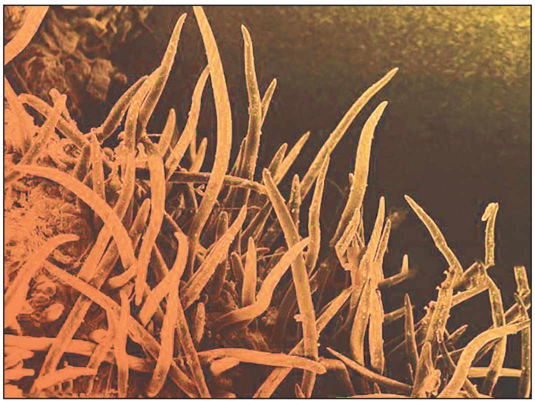
9. A peach is a fruit which has tiny hairs on its surface. A nectarine is a type of peach where the hairs are absent. The production of hair on the surface is controlled by a single gene. A dominant allele codes for the production of hair. A mutation to this gene produces a recessive allele which does not produce hair. **Image 9.1** is a picture produced by an electron microscope showing the hair growing on the surface of a peach.

Image 9.1



Peach scab disease is caused by the fungus *Cladosporium carpophilum* and can affect both peaches and nectarines. Fungi reproduce by producing spores, which are spread by wind and rain. When the spore lands on the surface of a fruit, the fungus begins to grow and scabs soon develop. Nectarines are more likely than peaches to develop scab disease. **Image 9.2** is a picture of a nectarine showing scabs caused by *Cladosporium carpophilum.*

Image 9.2

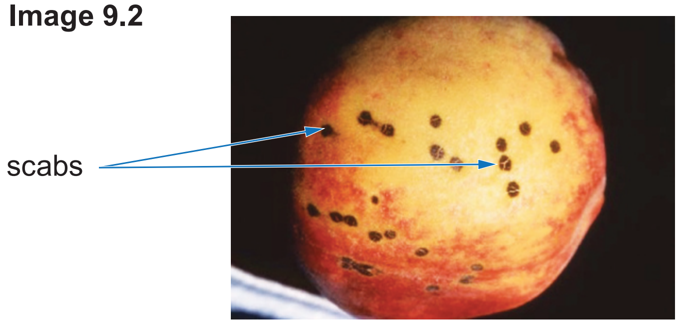
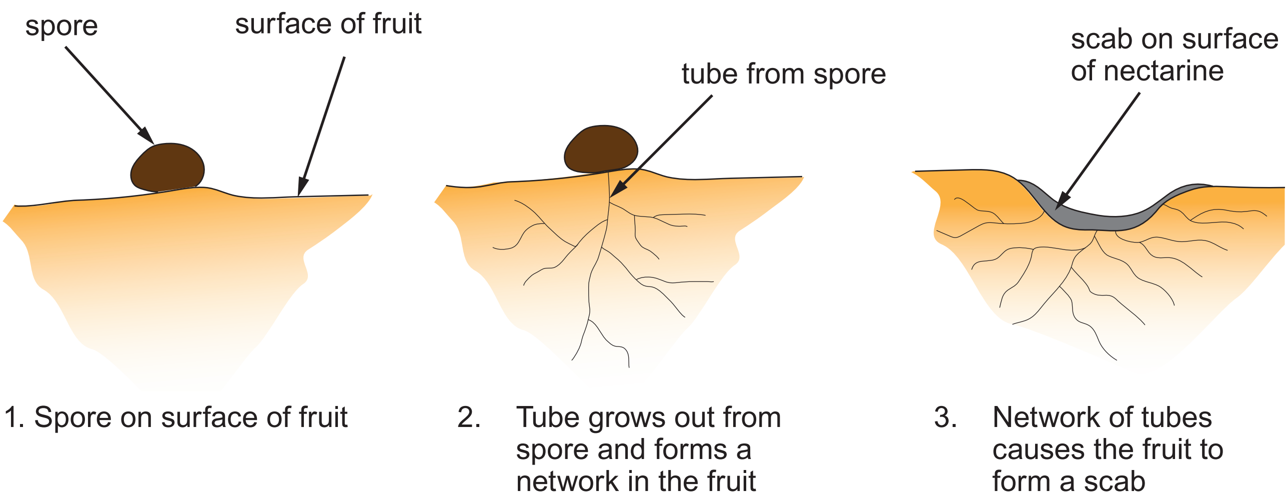


Image 9.3 shows a fungal spore developing and forming a scab.

Image 9.3



Examiner only

(a) (i) State why scientists use the name *Cladosporium carpophilum* rather than the common name peach scab when discussing this disease. [1]

....

....

(ii) Using all the information given, suggest why nectarines are more likely than peaches to develop scab disease. [2]

....

....

....

....

(b) (i) State what is meant by the term allele. [1]

....

....

(ii) State what is meant by a mutation. [1]

....

....

(iii) Give **one** example of an **environmental** factor which increases the rate of mutations. [1]

....

Continued overleaf

Examiner only

(c) (i) Use the letters **H** and **h** to show the result of a genetic cross between two peach trees that would result in the production of seeds that could grow into nectarine trees. [3]

Key: **H** = allele for hairy fruit

h = allele for hairless fruit

Phenotype = peach tree × peach tree

Genotype = ... × ....

Gametes		

(ii) Using your answer from the Punnett square above, state the probability of producing a nectarine tree from this cross. [1]

Probability = ...

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END OF PAPER